Development of Interactive 3D Imaging System for Hepatic Angiography

Essam A. Rashed, Ahmed M. Ghanem, Ahmad Amin, Ayman Atia, Mohammad al-Shatouri and Hiroyuki Kudo

***Abstract*—Egypt is witnessing over the next decade many challenges in the field of healthcare, especially with regard to the spread of hepatitis and coincides with the spread of liver hepatocellular carcinoma. Because most cases of liver cancer in Egypt are detected in very late stages, the use of surgical resection, liver transplantation and percutaneous ablative therapies constitutes unsuitable therapeutic options either due to high recurrence rate or unfeasibility. Therapy sessions can be made through the introduction of chemotherapy using a catheter directly into the hepatic artery supplying the tumor guided by angiography imaging system. This method of treatment known to prevent the patient from different problems associated with surgical treatment, but it is still needs to be further improved to maximize the benefits and minimize the risks. Hepatic angiography is an x-ray study of the blood vessels that supply the liver. The procedure uses a catheter that is placed into a blood vessel through a small incision. The catheter is guided using the x-ray images obtained through the interventional session. During angiography, hepatic arterial supply is usually displayed in one, two or three projections. Mental 3D interpretation of the anatomy is not an easy task. Reaching the target supply artery by the catheter tip is mandatory to obtain satisfactory tumor response and reduce complications and recurrence. This work aims to develop an interactive 3D imaging system of hepatic angiography. The developed system uses a set of 2D images measured over few view angles to reconstruct a full 3D volume of the hepatic arteries. The problem can be thought as a combination of three main approaches. (1) Image reconstruction of 3D artery volume from few number of projections (each is presented as 2D image), (2) automatic detection of the catheter roadmap to the labeled artery which feed the tumor, and (3) interactive system to control and display images using simple gestures of the physician.**

***Index Terms*—Image reconstruction, computed tomography, TACE, interactive control, ubiquitous computing, hepatic angiography**

I. Introduction

Egypt is witnessing over the next decade many challenges in the field of healthcare, especially with regard to the spread of hepatitis and coincides with the spread of liver hepatocellular carcinoma. Hepatocellular carcinoma (HCC) is one of the most common malignancy in Egypt with 100 to 1,200 confirmed cases per 1M, which represent the leading cause of death from other cancer sites [1]. About 5.9% of chronic liver disease (CLD) patients in Egypt are suffering from HCC. It is ranked the fourth among all types of cancers in Egypt. In cirrhotic, about 70% of liver related deaths are due to HCC. Egypt is ranked number one among Arab countries regarding the incidence of HCC [2]. The number of positive HCC cases in Egypt start growing since 1975 and expected to reach its peak around the year 2025 [3]. Because most cases of liver cancer in Egypt are detected in very late stages, the use of surgical resection, liver transplantation and percutaneous ablative therapies constitutes unsuitable therapeutic options either due to high recurrence rate or unfeasibility. Therefore, treatment using catheter through interventional radiology is highly preferred. Therapy sessions can be made through the introduction of chemotherapy using a catheter directly into the hepatic artery supplying the tumor guided by angiography imaging system well-known as transarterial chemoembolization (TACE). This method of treatment known to prevent the patient from different problems associated with surgical treatment, but it is still needs to be further improved to maximize the benefits and minimize the risks.

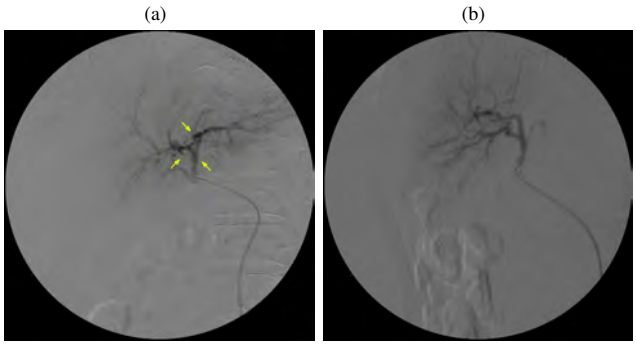

Fig. 1. A sample 2D DSA images for hepatic angiography obtained through TACE planning session. (a) and (b) are obtained at 75° and 0° angles, respectively. Arrows in (a) indicate the challenge in understanding the 3D anatomy and the deception of catheter roadmap for TACE. These conflict points becomes clear by looking from different view such as in (b).

Hepatic angiography is an x-ray study of the blood vessels that supply the liver. The procedure uses a catheter that is placed into a blood vessel through a small incision. The catheter is guided using the x-ray images obtained through the interventional session, see Fig. 1 for a sample hepatic angiographic image. During TACE, hepatic arterial supply is usually displayed in one, two or three projections. Common clinical routine include the using of anteroposterior and 15

This work is partially supported by the Applied Scientific Research Projects (ASRP) program, Ministry of Higher Education, Egypt under Grant number ASRP1-008-SUE.

E.A.Rashed is with the Department of Mathematics, Faculty of Science, Suez Canal University, Ismailia 41522, Egypt (e-mail: erashed@science.suez.edu.eg). A.M.Ghanem is with the Department of Electrical Engineering, Faculty of Engineering, Suez Canal University. A.Amin and A.Atia are with the HCI-LAB, Department of Computer Science, Faculty of Computers and Information Systems, Helwan University, Egypt. M.al-Shatouri is with the Department of Radiology, Faculty of Medicine, Suez Canal University, H.Kudo is with the Department of Computer Science, Faculty of Engineering, Information and Systems, University of Tsukuba, Tsukuba, Japan.

978-1-4799-0534-8/13/$31.00 ©2013 IEEE

degree Rt. anterior oblique projections only. Mental 3D interpretation of the anatomy during the TACE treatment session is not an easy task. Reaching the target supply artery by the catheter tip is mandatory to obtain satisfactory tumor response and reduce complications and recurrence. Objective 3D image prior to chemotherapy and embolization would have been beneficial because of the followings. (1) Provide the necessary information required to choose the suitable single catheter rather than trying multiple catheters which increasing the cost of the procedure. (2) Shorten the time of the procedure, machine usage and consequently reduce radiation dose given to patient and staff. (3) Minimize the procedure failure rate. (4) Reduce the usage of microcathers that double the cost of the procedure. (5) Improve quality of targeting the injected drug which increasing the procedure efficacy, and (6) avoid the arteries supplying the parenchyma, thus reducing post procedure risk of major complications (15%), deaths (2%) and reduction of vitality aspect of quality of life. One of the most important problems faced by the physician in the application under consideration is how to develop a 3D perception of hepatic blood vessels anatomy through small number of 2D images (less than 10) provided by a conventional C-arm machine. Moreover, how to determine the artery supplying the tumor accurately and quickly.

II. Methods

This work aims to develop an interactive 3D imaging system of hepatic angiography. The developed system uses a set of 2D images measured over a small number of views. The problem can be thought as a combination of three main approaches. (1) image reconstruction of 3D artery volume from a small number of projection views (each is presented as 2D image), (2) automatic detection of the catheter roadmap to the labeled artery which feed the tumor, (3) interactive system to control and display images using simple gestures of the physician. As this is the first paper to report this work, results presented here are limited to the image reconstruction and the interactive control only.

A. Image reconstruction

In x-ray computed tomography (CT), the image reconstruction problem can be formulated to solve the following system of linear equations

$$A\mathbf{x} = \mathbf{b}, \tag{1}$$

where $\mathbf{x} = (x_1, \ldots, x_n)^\top$ is a vector represent the object image in a discrete form with n pixels, $\mathbf{b} = (b_1, \ldots, b_m)^\top$ is a vector of m measured projection data and $A = \{a_{ij}\}$ is the $m \times n$ system matrix. A single element a_{ij} of matrix A represents the physical and statistical properties that defines how image pixel x_j contributes to the projection data measured at the detector bin b_i. Considering that $\mathbf{x}$ is sparse in the target application of blood vessels imaging [4], one potential method to solve Eq. (1) is the equality constrained ℓ_1 minimization problem, known as basic pursuit

$$\text{minimize} \quad \|\mathbf{x}\|_1 \quad \text{subject to} \quad A\mathbf{x} = \mathbf{b}. \tag{2}$$

The solution of problem in Eq. (2) can be achieved with several approaches. The alternating direction method of multipliers (ADMM) was introduced early in the field of convex optimization [5]. The method was formulated to improve the convergence propertied of the Method of Multipliers (MM) with rather relaxed constraint. In ADMM form, basic pursuit problem in Eq. (2) can be written as:

$$\text{minimize} \quad f(\mathbf{x}) + \|\mathbf{u}\|_1 \quad \text{subject to} \quad \mathbf{x} - \mathbf{u} = \mathbf{0}, \tag{3}$$

where f is the indicator function of $\{\mathbf{x} \in \mathcal{R} \mid A\mathbf{x} = \mathbf{b}\}$. The ADMM algorithm is performed through the following steps:

$$\mathbf{x}^{k+1} = \Pi(\mathbf{u}^k - \lambda^k) \tag{4}$$

$$\mathbf{u}^{k+1} = \text{SoftThres}(1/\rho, \mathbf{x}^{k+1} + \mathbf{u}^k) \tag{5}$$

$$\lambda^{k+1} = \lambda^k + \mathbf{x}^{k+1} - \mathbf{u}^{k+1} \tag{6}$$

where ρ is a penalty parameter used as the step size, k is the iteration number, Π is projection onto $\{\mathbf{x} \in \mathcal{R} \mid A\mathbf{x} = \mathbf{b}\}$ and SoftThres is the soft thresholding operator defined as:

$$\text{SoftThres}(t, a) = \begin{cases} a - t & a > t \\ a + t & a < -t \\ 0 & |a| \leq t \end{cases}. \tag{7}$$

The ADMM algorithm can be thought as three main steps. First, enforcement of data fidelity term as in Eq. (4) followed by a soft-thresholding operation (Eq. (5)) and finally a parameter update step as in Eq. (6). In results presented in this paper, Eq. (4) is satisfied using the well-known Algebraic Reconstruction Technique (ART) algorithm [6]. The interesting behavior of this algorithm is that in practical implementation, it is enough to use a single local iteration of data fidelity enforcement term at each global iteration k, to obtain a high-quality image. The theoretical explanation of this behavior is yet unclear. The proposed iterative reconstruction method is of potential useful for the target application, providing quick, effective and easy-to-implement reconstruction method. The computation cost is almost the same as several conventional methods of the Projections Onto Convex Sets (POCS) family.

B. Catheter road-mapping

The second stage in the proposed system is the automatic detection of the catheter roadmap to the labeled artery feeding the tumor. To provide such roadmap, we reconstruct a tree that represents the patient hepatic arteries network, find the road in this tree to the tumor, then map the discovered road on the tree to the reconstructed 3D volume of the hepatic arteries. Figure 2 shows the block diagram of the catheter roadmap generation stage.

As shown in Fig. 2, the first step in the block diagram is extracting the 3D skeleton. We use a modified version of algorithm in [7] to extract the skeleton. In such modified version, distance transformation is calculated for the 3D volume of the hepatic arteries, the skeleton is extracted as in [7], and finally, a post processing step is applied to ensure that the skeleton is connected and have one pixel thickness. After the extraction process, the skeleton is pruned to eliminate the false

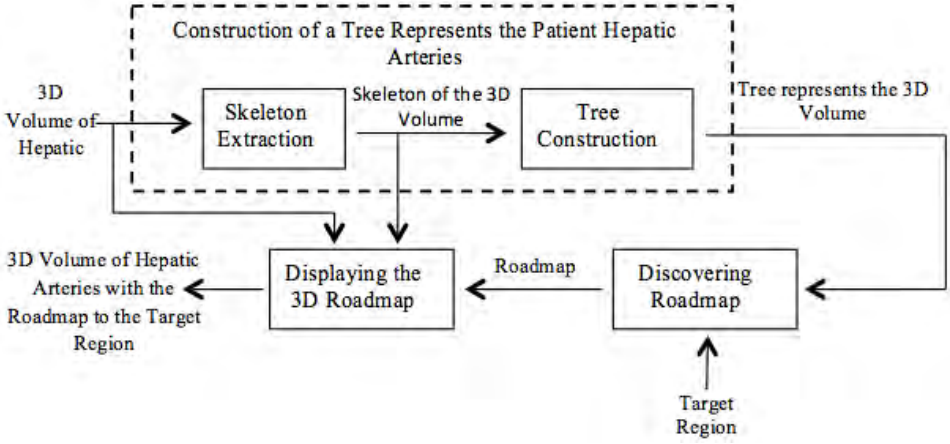

Fig. 2. Block diagram of the automatic detection of the catheter roadmap to the labeled artery which feed the tumor.

branches: the branches that have one end point and length less than certain threshold, which is determined according to the image resolution and the level of required details in the constructed tree. Then, each point in the skeleton is classified as branch point, edge point, or end point according to the number of neighbor points. Within the 26-neighbor cube, the end point has only one neighbor point; the edge point has two neighbor points while the branch point has three or more neighbor points. Thus, the tree is constructed where the branch and end points are considered as nodes.

Following the tree construction, the roadmap discovering step begins by allowing the physician to identify the target region. The nearest node to the target region is identified and labeled as target node. Using the depth first searching technique, the roadmap between the catheter position and the target node is discovered and represented as a set of nodes: the first node is the nearest one to the catheter while the ending node is the nearest one to the target region. Using this set of nodes, the displaying step; final step, identifies all edge points along the roadmap and labels them on the skeleton. Then all nodes and edge points are labeled on the reconstructed 3D volume of the hepatic arteries. These roadmap points represent the centerline of the hepatic arteries that construct the path to the target region. To display the roadmap on the 3D volume, we need to extract all the points along the path to the target region not just the centerline points. Therefore, we use the region growing technique with stopping criteria based on the distance between a point and its nearest roadmap point. Such distance is determined using the maximum value in the 3D volume distance transformation which is calculated previously in the first step.

C. Volume rendering

After the 3D volumetric shape is generated the next step is to start rendering it to be manipulated later, in this step we are doing the entire rendering using XNA 4.0 framework. We used volume ray casting technique to render the 3D volume which is basically rays being emitted at the volume at equidistant spaces, as this ray marches through the volume, scalar values of this volume are mapped accordingly, through this mapping, RGBA colour value is produced to the point that is being sampled by the ray. After sampling all the points it results in rendering the whole 3D volume to be then ready for manipulation. This is implemented by surrounding the volume texture with a box, rendering this box, and sampling the texture. We render the front and back of the box facing the texture and then we sample the texture to find the sampling ray. Then iteratively sample the volume by advancing the current sampling position along the ray at equidistant steps. Finally, we use front-to-back compositing to accumulate the pixel colour.

D. Interactive display control system

We propose an interactive system that can allow physicians to interact with the reconstructed 3D image using gestures. We study the appropriate input method for physicians while their hands are not completely free. A depth camera will be used to collect data that will be the input of a hand-tracking system that detects the physicians hands, then we track the gestures made by detected hands to rotate, zoom and flip the 3D images. We study number of predefined and customised gestures for each physician to manipulate the images. Figure 3 illustrates the proposed system overview. The 3D images will be displayed on a large display screen to show feedback. The main challenge in this work is to find the appropriate gestures for controlling images and to distinguish between intended and unintended gestures for interaction.

Gallo *et al.* discussed a controller-free exploration of medical image data [8]. They have developed an interactive system to control the display of medical images using Microsoft Kinect [9] as the only input device using hand and arm gestures of the user, and since it is a touch-free interaction it is suitable for use in operating rooms, where non-sterilizable devices cannot be used. However, this system needed a calibration phase which took around 30 seconds in order to be able to complete the skeleton fitting process to compute the arm length of the user, identify the dominant hand, and other parameters setup. They propose a new way

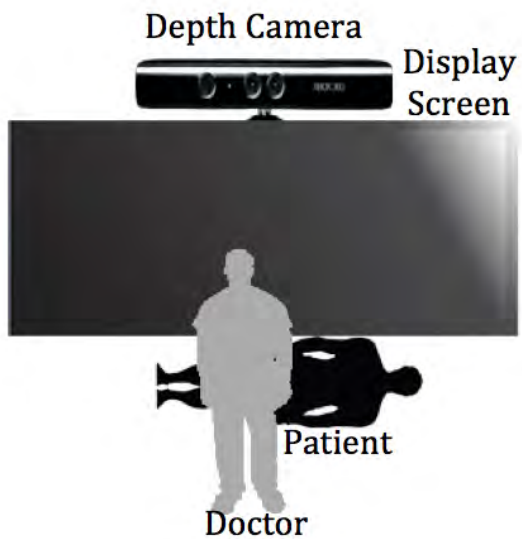

Fig. 3. The proposed interactive system overview.

to distinguish between the intended and the non-intended gestures by defining an activation area. The user's arm must be outstretched more than 55% of its length. However, this system has gestures that must be defined using both hands like the zoom gesture, it involves keeping both hands with open palms inside the activation area with the open palms moving in opposite direction.

In the proposed system we aimed at reaching a calibration time in about 3 seconds, proposed a new way to distinguish between intended and non-intended gestures, also propose a novel way to handle multiple users access and controlling 3D images display using only a single hand gestures.

1) Proposed Interactive system versus conventional input methods: This system was developed to reduce the time of interaction used by the physician during TACE procedure. The controller is usually placed at the end of the operation table so if the physician needs to manipulate the display, he has to leave the patient and go to the controller to perform the required manipulation process. This activity is time consuming and there is a possibility that the physician miss-click the desired button as all the buttons are covered with sterilized sheet so the buttons labels are not clearly visible. Furthermore, the main control unit is outside the operation room and any advanced manipulation requires the surgeon to request an assistant in the control room to perform a specific task. This process is also a time consuming with potential of faulty response. In the proposed system we aim at reducing the time taken and human errors to manipulate the display by making the physician perform single hand gestures during the operation without leaving the position near by the patient.

2) Interaction details: The physician should make a push gesture (moving his hands forward and backward) to start controlling the 3D image in steady mode. If another person interfere the scene, the system will not follow the interfering gestures unless the interfering person perform another push gesture. If the physician wants to command the interface to

Fig. 4. Different interaction gestures, the push gesture (left), the grip pose moving horizontally and vertically to rotate the volume (middle) and the grip pose moving towards the Kinect to zoom in and backwards to zoom out (right).

start moving/zooming the rendered 3D image, he/she must start his gesture with a grip pose. The physician have to move in some direction, hence we call it the interaction mode. The rotate gesture is performed by moving the hand horizontally right or left or vertically up and down to rotate the volume in different directions. To zoom in, the physician should also use the Grip Pose and push his hand towards the depth camera, or away from the depth camera to zoom out. Figure 4 illustrate the interaction gestures performed by the system user. If another user wants to take control of the system while another one is using it, then the new user is requested to perform the push gesture with his hand open, then the control transfers from the old user to the new one.

III. Results and Discussion

A. Image reconstruction

Experimental studies are implemented using FORBILD lung phantom [10], shown in Fig. 5(a). The volume size is $256\times256\times256$ pixels and the projection data is measured using detector size of 256×256 over 5 and 8 projections with different spacing angles. We consider three data acquisition setups. First, we acquire 8 projections with angle spacing of 22.5°. Second, we reduce the number of projection to only 5 projections with spacing angle 36.0° and finally we fix the number of projections and reduce the spacing angle to 30.0°. This setup is shown in Fig. 5 (c), (g) and (h). Reconstruction is implemented using conventional Algebraic Reconstruction Technique (ART) and compared with the developed method. Results obtained from different imaging setups are shown in Fig. 5. Although the number of projections is extremely small with scanning orbit less than short-scan, the reconstructed volume is almost accurate even with small size vessels. The number of iterations was fixed to only 50 iterations which is computed using 261, 166 and 164 seconds in the three predefined setups, respectively. We have used a machine of 1.8 GHz Core i5 processor and 4.0 GB memory. Reconstruction images provide interesting results especially when the number of projection views is very small and the scanning orbit is less than 180°, which would provide potential privilege to effectively reduce procedure time and consequently the patient dose.

B. Interactive control

To evaluate the interactive control system, shown in Fig. 6, we conducted an experiment on six users aged from 20 to 60

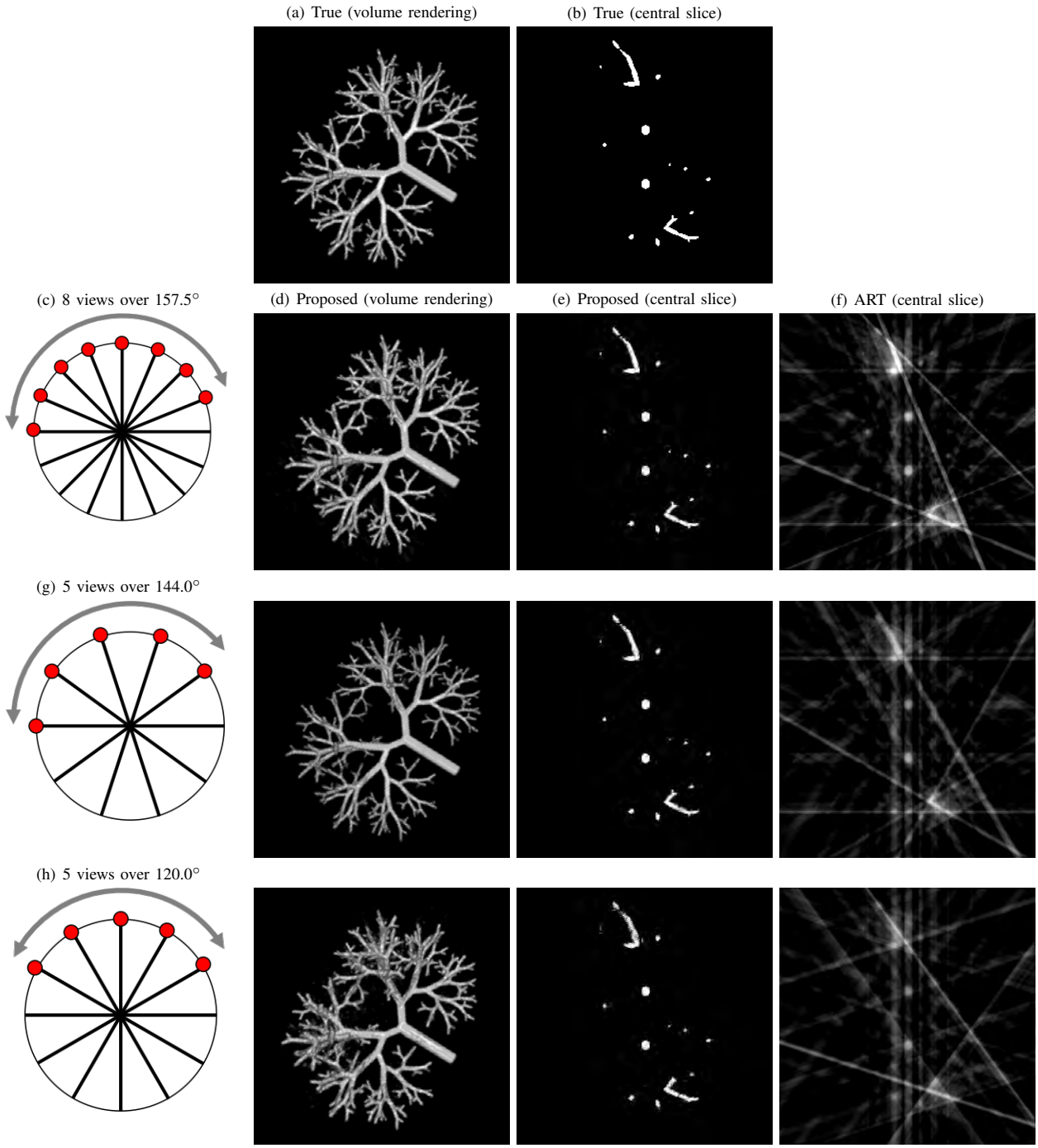

Fig. 5. Image reconstruction results using different setups. (a) is the true phantom with corresponding central slice in (b). Reconstruction in rows 2-4 are acquired from imaging setup shown in (c), (g) and (h) respectively.

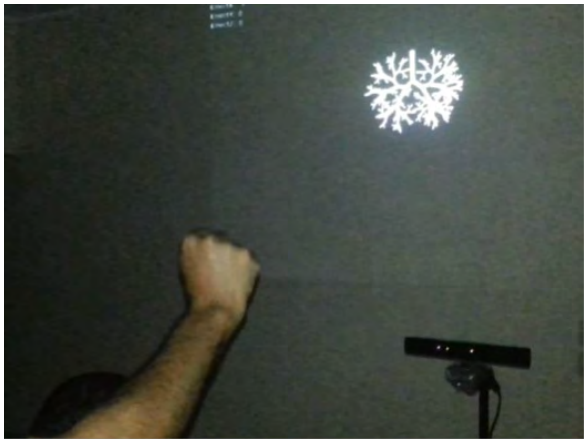

Fig. 6. A user is using the Grip Pose to interact with the 3D volume rendering display of the phantom using Kinect device.

years old. Users were asked to enter a sequence of gestures. Start interaction mode (steady state), move left, right, up, down, zoom in and zoom out. Each user performed the whole sequence for five times with a total of thirty gestures per each user. In this experiment, we aimed to measure two factors, the accuracy of this interaction and the time to complete the pre-set scenario of gestures. The interaction accuracy results showed that minimum average accuracy was 90% and maximum average accuracy reached 100%. Figure 7 shows the users accuracy and time required to perform the interaction scenario for different test sample. The interaction time results showed that maximum average time was 23 seconds and minimum average time was 13.6 seconds.

IV. Conclusion

The aim of this work is to develop a complete system for interactive 3D imaging of hepatic angiography during TACE procedure. This include a high-speed 3D image reconstruction from small number of projections, automatic detection of the catheter roadmap and interactive control of image display. The developed interactive system provide a potential useful approach that fit with the target imaging modality.

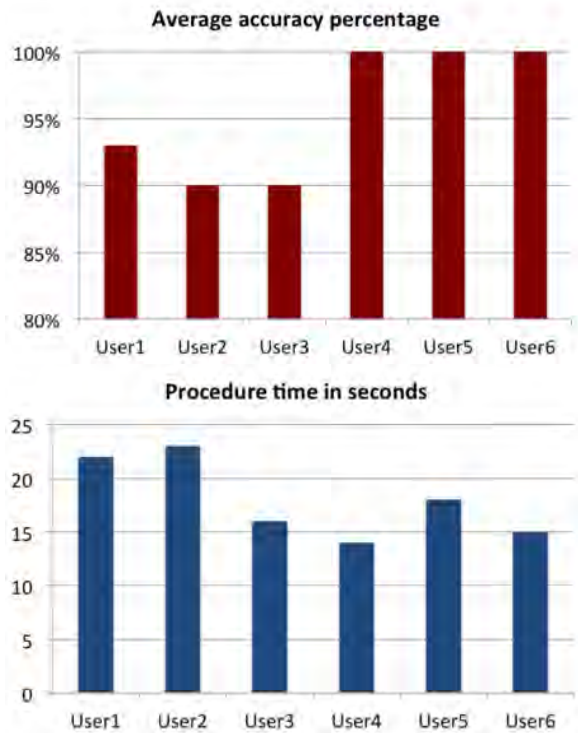

Fig. 7. Evaluation results of the interactive control system for different users. Top is the accuracy results and bottom is the time used to perform the whole interaction sequence scenario.

References

[1] A. R. El-Zayadi, *et al.*, "Effect of surveillance for hepatocellular carcinoma on tumor staging and treatment decisions in Egyptian patients," *Hepatol. Int.*, vol. 4, pp. 500-506, 2010

[2] E. M. Lehman, *et al.*, "Patterns of hepatocellular carcinoma incidence in Egypt from a population-based cancer registry," *Hepatology Research*, vol. 38, pp. 465-473, 2008

[3] D. B. Rein, *et al.*, "Forecasting the morbidity and mortality associated with prevalent cases of pre-cirrhotic chronic hepatitis C in the United States," *Dig. Liver Dis.*, vol. 43, pp. 66-72, 2011

[4] M. Li, H. Yang and H. Kudo, "An accurate iterative reconstruction algorithm for sparse objects: application to 3D blood-vessel reconstruction from limited number of projections," *Phys. Med. Biol.*, vol. 47, pp. 2599-2609, 2002

[5] D. Gabay and B. Mercier, "A dual algorithm for the solution of nonlinear variational problems via finite element approximation," *Comp. Math. Appl.*, vol. 2, pp. 17-40, 1976

[6] G. T. Herman and L. B. Meyer, "Algebraic reconstruction techniques can be made computationally efficient," *IEEE Trans. Med. Imaging*, vol. 12, pp. 600-609, 1993

[7] K. Palàgyi, J. Tschirren, E. A. Hoffman and M. Sonka, "Quantitative analysis of pulmonary airway tree structures," *Computers in Biology and Medicine*, vol. 36, pp. 974-996, 2006

[8] L. N. Gallo, A. P. Placitelli, and M. Ciampi. "Controller-free exploration of medical image data: Experiencing the kinect," *In International Symposium on Computer-Based Medical Systems*, pp. 1-6, 2011

[9] Microsoft Corporation. XBox Kinect. http://www.xbox.com/kinect

[10] http://www.imp.uni-erlangen.de/phantoms/lung/lung.html